

Evidence review

# The harness (still) matters

Public studies show that the coding-agent harness changes performance, cost, token use and runtime. They do not identify one harness that wins across every model and task.

Evidence captured on 12 July 2026

# 13 benchmarks, ordered by how much they can tell us

The evidence is consistent on one point: changing the harness can change quality, cost and speed. It is not consistent on which harness wins.

**Harnesses are not neutral**  
Tools, context, retries and timeouts change the evaluated system.

**The ranking changes**  
No harness wins across every model and task.

**Test the combination**  
Evaluate model, harness, task and budget together.

## 1 Operational and broad evidence

Start here: larger, more operational or more extensively reported comparisons.

|    |                                   |
|----|-----------------------------------|
| 01 | Databricks coding-agent benchmark |
| 02 | Warden benchmark                  |
| 03 | Coding Agent Index                |
| 04 | DrugDiscoveryBench                |
| 05 | Agent Security League             |

## 2 Focused domain evidence

Strong task-specific evidence, but conclusions may not transfer outside the domain.

|    |                           |
|----|---------------------------|
| 06 | scBench-Long              |
| 07 | Vulnerability-Agent-Bench |
| 08 | WildClawBench             |
| 09 | CORE-Bench Hard           |

## 3 Small and diagnostic evidence

Useful signals with limited task counts or no direct measure of coding quality.

|    |                |
|----|----------------|
| 10 | HarnessBench   |
| 11 | A-CODE-CLI     |
| 12 | OpenBench M4.5 |
| 13 | Harness Tax    |

Databricks evaluated recent internal pull requests: Pi used about one-third as much context, and equal-quality runs sometimes differed by more than 2 times in cost.

What they did

Databricks replayed recent human-written pull requests from its multi-million-line internal codebase. Engineers reviewed whether each agent's changes passed repository tests and matched the intended change.

|                 |                                        |
|-----------------|----------------------------------------|
| Tasks           | Not published                          |
| Repeated trials | not clear                              |
| Grader          | Repository tests reviewed by engineers |

What we observe

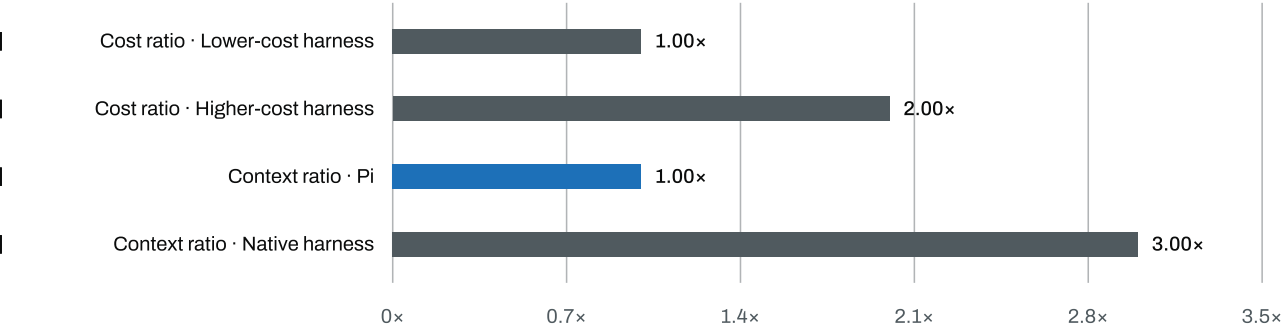
Pi used about one-third as much context per turn. Some equal-quality matched runs differed by more than 2 times in cost.

Limit

The source does not publish task count, matched rows, run-level data or uncertainty.

Databricks results

Published results



The source reports 'more than 2 times' for cost and 'about 3 times' for context. These are directional ratios, not exact run-level values.

# Sentry tested 86 known vulnerabilities: Pi found a similar or larger share while costing about one-quarter to one-fifth as much as Claude SDK.

## What they did

Sentry gave agents files connected to 86 historical vulnerabilities and measured how many known findings they recovered. The published table compares matched Claude models running in Pi and Claude SDK.

|                 |                     |
|-----------------|---------------------|
| Tasks           | 86                  |
| Repeated trials | no                  |
| Grader          | Known-corpus recall |

## What we observe

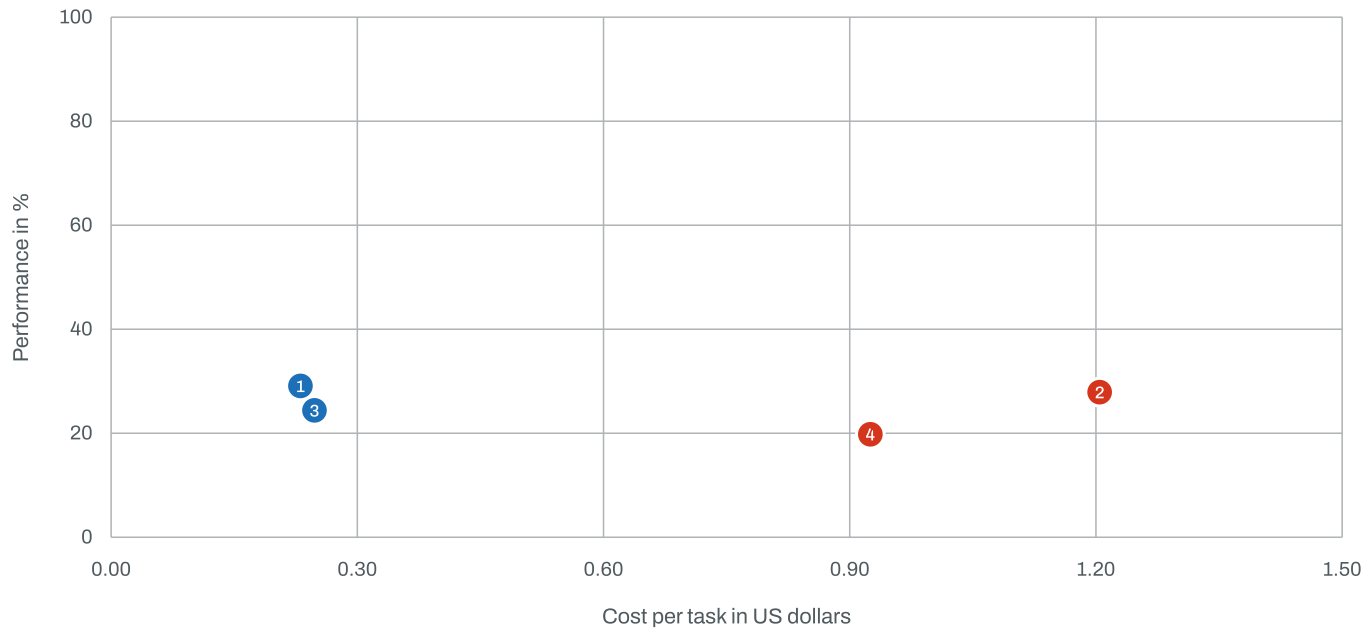
Pi cost less and found a similar or larger share of the known corpus for the matched Claude models.

### Limit

The comparator is Claude SDK, not the Claude Code command-line product. The benchmark does not measure whole-repository discovery.

Sentry results  
Dataset or repository

## Published results



| Point | Model             | Harness    | Result · cost  |
|-------|-------------------|------------|----------------|
| 1     | Claude Sonnet 4.6 | Pi         | 29.1% · \$0.23 |
| 2     | Claude Sonnet 4.6 | Claude SDK | 27.9% · \$1.20 |
| 3     | Claude Opus 4.8   | Pi         | 24.4% · \$0.25 |
| 4     | Claude Opus 4.8   | Claude SDK | 19.8% · \$0.93 |

# Artificial Analysis combined 321 coding tasks: OpenCode scored higher with Opus 4.7, while Claude Code was cheaper and faster.

## What they did

Artificial Analysis combines 321 tasks from DeepSWE, Terminal-Bench v2 and SWE-Atlas-QnA. It repeats each configuration and reports a composite score alongside cost and runtime.

|                 |                                                |
|-----------------|------------------------------------------------|
| Tasks           | 321                                            |
| Repeated trials | yes                                            |
| Grader          | Program checks, test suites and rubric scoring |

## What we observe

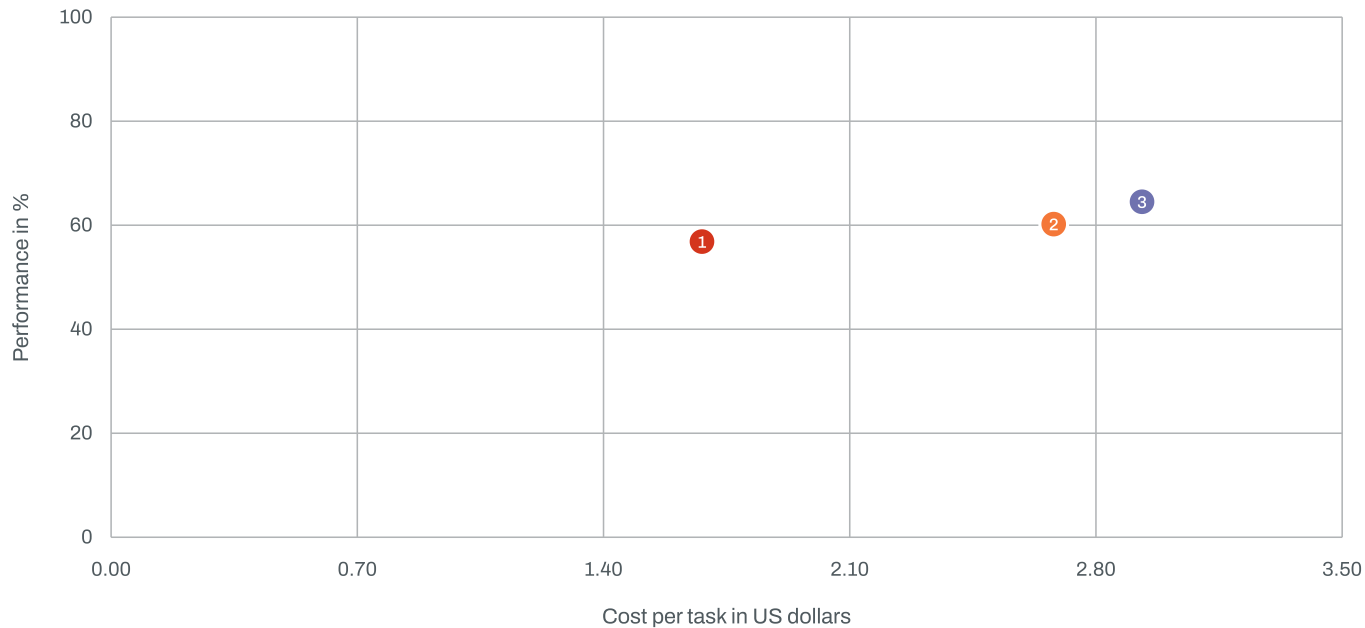
OpenCode led the captured Opus 4.7 medium comparison. Claude Code was cheaper and faster.

### Limit

The index averages 3 different benchmark families. The live results and harness versions can change.

Artificial Analysis results

## Published results



| Point | Model           | Harness     | Result · cost  |
|-------|-----------------|-------------|----------------|
| 1     | Claude Opus 4.7 | Claude Code | 56.9% · \$1.68 |
| 2     | Claude Opus 4.7 | Cursor CLI  | 60.2% · \$2.68 |
| 3     | Claude Opus 4.7 | OpenCode    | 64.5% · \$2.93 |

## DrugDiscoveryBench repeated 82 expert workflows: mini-SWE-agent led the matched GPT-5.5 comparison, while Codex placed in the middle.

### What they did

Scale AI ran 82 expert-curated drug-discovery and life-sciences workflows 3 times. An LLM judge scored each result against expert-authored rubrics and reference answers.

|                 |                                                |
|-----------------|------------------------------------------------|
| Tasks           | 82                                             |
| Repeated trials | yes                                            |
| Grader          | Expert-authored rubrics scored by an LLM judge |

### What we observe

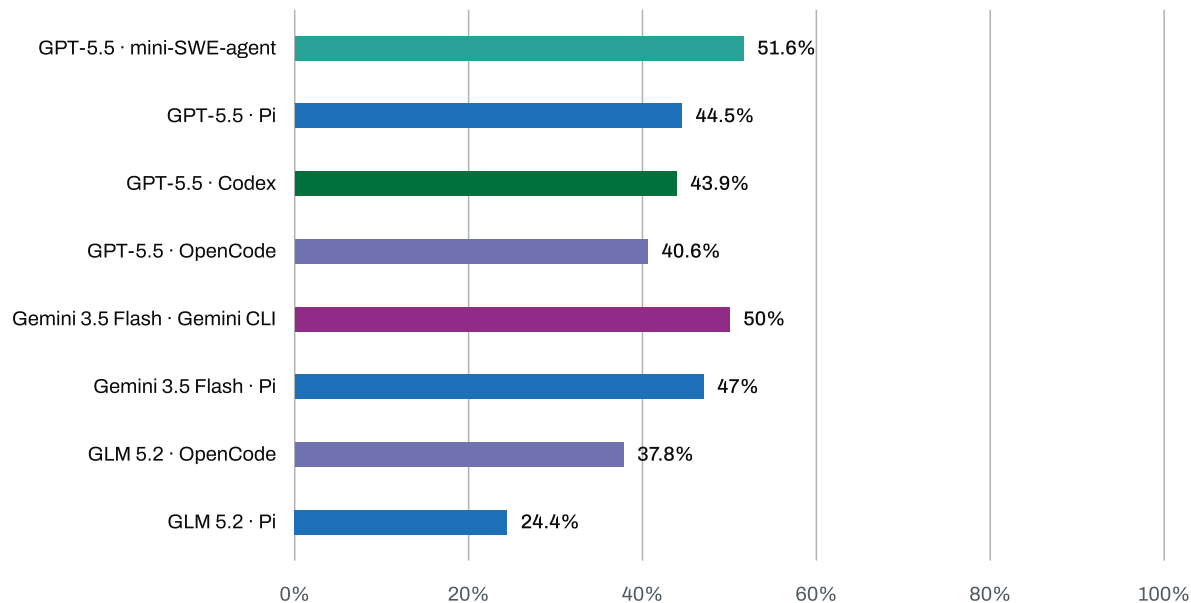
GPT-5.5 performed best in mini-SWE-agent. Codex was neither the best nor worst harness for that model.

#### Limit

The paper plots cost but does not publish a table of exact cost points. The outcome depends on an LLM judge.

Scale AI results  
Dataset or repository

### Published results



# Endor Labs tested 200 vulnerability repairs: Cursor led for Opus, but Claude Code led for Sonnet 5.

## What they did

Endor Labs asked each configuration to repair 200 real vulnerabilities. Hidden tests separately measured whether the software still worked and whether the vulnerability was fixed.

|                 |                                      |
|-----------------|--------------------------------------|
| Tasks           | 200                                  |
| Repeated trials | no                                   |
| Grader          | Hidden functional and security tests |

## What we observe

Cursor led Claude Code for Opus 4.7 and 4.8. Claude Code led Cursor for Sonnet 5.

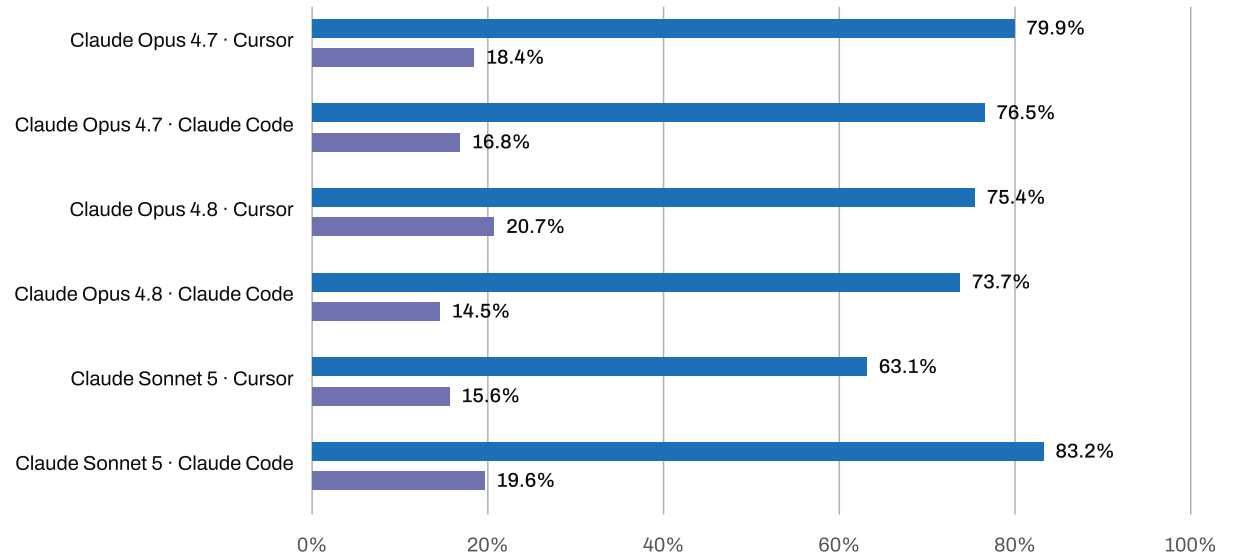
### Limit

There are no repeated trials or task-level costs. Timeouts explain part of the reversal.

Endor Labs results

## Published results

Functional pass rate Security pass rate



## scBench-Long repeated 21 biological workflows: the apparent harness differences are smaller than the reported uncertainty.

### What they did

LatchBio ran 21 long biological analysis workflows 3 times for each model and harness combination. Deterministic biological graders checked whether the analyses reached the required outputs.

|                 |                                  |
|-----------------|----------------------------------|
| Tasks           | 21                               |
| Repeated trials | yes                              |
| Grader          | Deterministic biological graders |

### What we observe

Claude Code led Pi on Opus 4.8 while Codex and Pi tied on GPT-5.5. The reported intervals overlap widely.

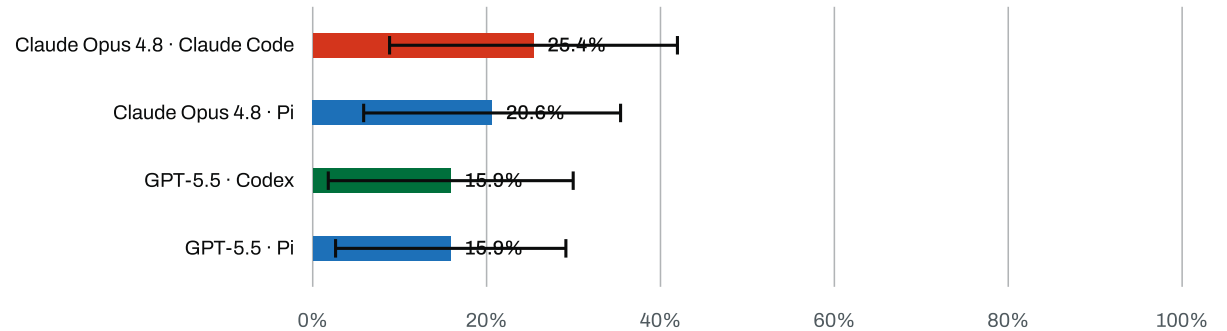
#### Limit

The public summary does not expose cost. The confidence intervals are wide.

LatchBio results

Dataset or repository

### Published results



# XOR tested 136 vulnerability repairs: Claude Code and Codex beat OpenCode on all 3 published matched-model pairs.

## What they did

XOR ran agents in isolated containers on real vulnerability-repair tasks. Its public explorer reports pass rate and cost for matched model and harness configurations.

|                 |                             |
|-----------------|-----------------------------|
| Tasks           | 136                         |
| Repeated trials | no                          |
| Grader          | Vulnerability repair checks |

## What we observe

Claude Code and Codex led the corresponding OpenCode configurations on the published matched pairs.

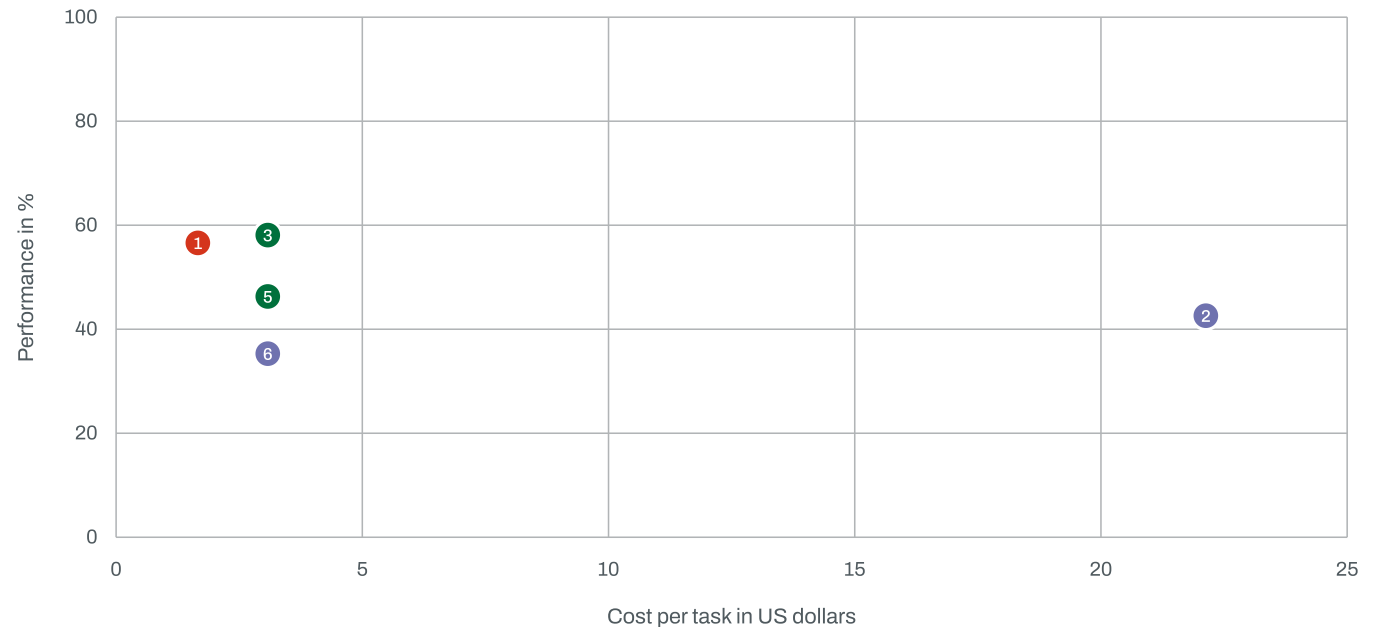
### Limit

The site alternates between 128 and 136 samples. Some reported cost differences need more explanation.

## XOR results

Dataset or repository

## Published results



| Point | Model           | Harness     | Result · cost   |
|-------|-----------------|-------------|-----------------|
| 1     | Claude Opus 4.6 | Claude Code | 56.6% · \$1.66  |
| 2     | Claude Opus 4.6 | OpenCode    | 42.6% · \$22.13 |
| 3     | GPT-5.2         | Codex       | 58.1% · \$3.08  |
| 4     | GPT-5.2         | OpenCode    | 46.3% · \$3.08  |
| 5     | GPT-5.2-Codex   | Codex       | 46.3% · \$3.08  |
| 6     | GPT-5.2-Codex   | OpenCode    | 35.3% · \$3.08  |

# WildClawBench tested 60 long-horizon tasks: changing the harness produced double-digit score shifts and changed the winner by model.

## What they did

InternLM evaluated 60 bilingual, multimodal and long-horizon tasks in harness-specific container images. Deterministic checks, environment checks and model judges contributed to the overall score.

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| Tasks           | 60                                                        |
| Repeated trials | not clear                                                 |
| Grader          | Deterministic checks, environment checks and model judges |

## What we observe

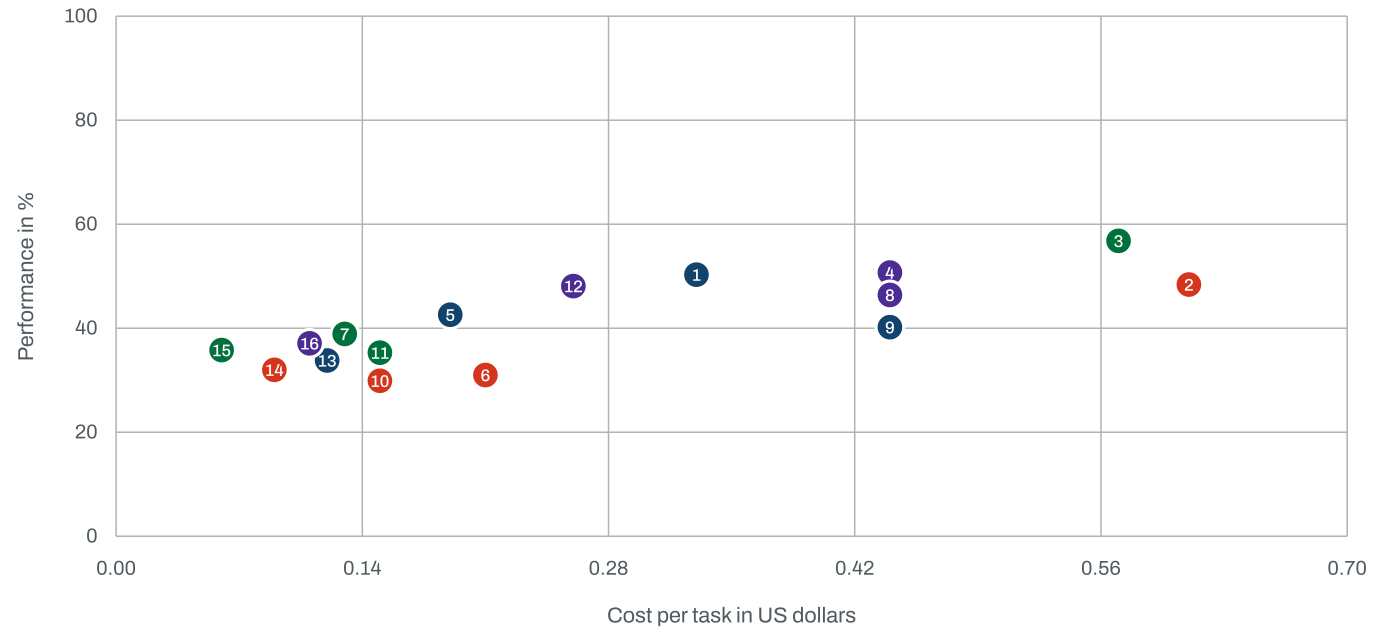
The best harness changed with the model. The source reports double-digit score changes from switching harness.

### Limit

Each harness image has different tools and prompt conventions. Repeat counts are not clear in the headline table.

InternLM results  
Dataset or repository

## Published results



| Point | Model   | Harness      | Result · cost  | Point | Model        | Harness      | Result · cost  |
|-------|---------|--------------|----------------|-------|--------------|--------------|----------------|
| 1     | GPT-5.4 | OpenClaw     | 50.3% · \$0.33 | 9     | MiMo V2 Pro  | OpenClaw     | 40.2% · \$0.44 |
| 2     | GPT-5.4 | Claude Code  | 48.4% · \$0.61 | 10    | MiMo V2 Pro  | Claude Code  | 29.9% · \$0.15 |
| 3     | GPT-5.4 | Codex        | 56.8% · \$0.57 | 11    | MiMo V2 Pro  | Codex        | 35.3% · \$0.15 |
| 4     | GPT-5.4 | Hermes Agent | 50.7% · \$0.44 | 12    | MiMo V2 Pro  | Hermes Agent | 48.1% · \$0.26 |
| 5     | GLM 5   | OpenClaw     | 42.6% · \$0.19 | 13    | MiniMax M2.7 | OpenClaw     | 33.8% · \$0.12 |
| 6     | GLM 5   | Claude Code  | 31% · \$0.21   | 14    | MiniMax M2.7 | Claude Code  | 32% · \$0.09   |
| 7     | GLM 5   | Codex        | 38.9% · \$0.13 | 15    | MiniMax M2.7 | Codex        | 35.8% · \$0.06 |
| 8     | GLM 5   | Hermes Agent | 46.4% · \$0.44 | 16    | MiniMax M2.7 | Hermes Agent | 37.1% · \$0.11 |

# CORE-Bench Hard tested 45 scientific reproductions: Claude Code more than doubled verified success while costing less than CORE-Agent.

## What they did

Princeton HAL asked agents to reproduce 45 research artefacts. The leaderboard verifies whether each system recreated the required scientific result and reports total cost.

|                 |                                |
|-----------------|--------------------------------|
| Tasks           | 45                             |
| Repeated trials | no                             |
| Grader          | Research artifact reproduction |

## What we observe

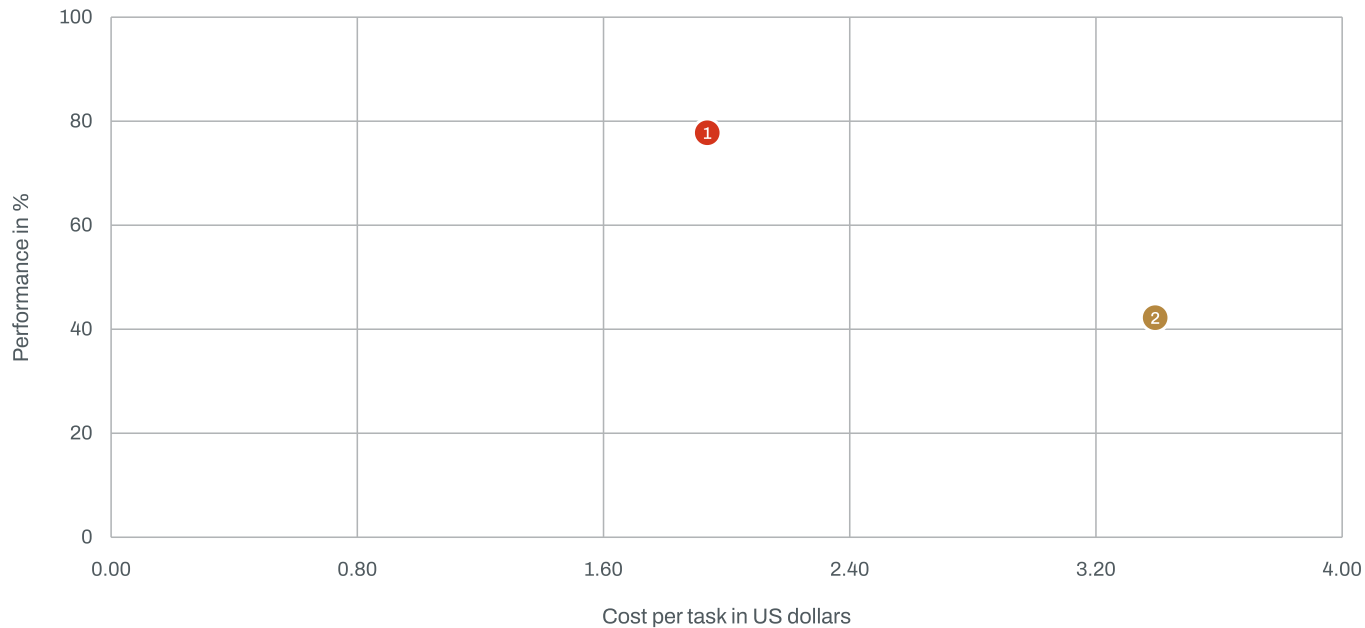
Claude Code more than doubled the verified success of CORE-Agent for Claude Opus 4.5 while costing less.

### Limit

There is one run per configuration. The systems differ across retrieval, tools and context policy.

Princeton HAL results  
Dataset or repository

## Published results



| Point | Model           | Harness     | Result · cost  |
|-------|-----------------|-------------|----------------|
| 1     | Claude Opus 4.5 | Claude Code | 77.8% · \$1.94 |
| 2     | Claude Opus 4.5 | CORE-Agent  | 42.2% · \$3.39 |

## HarnessBench tested 27 recent bug fixes: several configurations finished within one task of the leader, but runtime differed more clearly.

### What they did

HarnessBench selected 27 recent bug-fix issues from 9 open-source repositories. Hidden tests checked the core fix and regressions, with one run for each configuration.

|                 |                                  |
|-----------------|----------------------------------|
| Tasks           | 27                               |
| Repeated trials | no                               |
| Grader          | Hidden core and regression tests |

### What we observe

Several conditions finished within one task of the observed leader. Runtime differences were clearer than pass-rate differences.

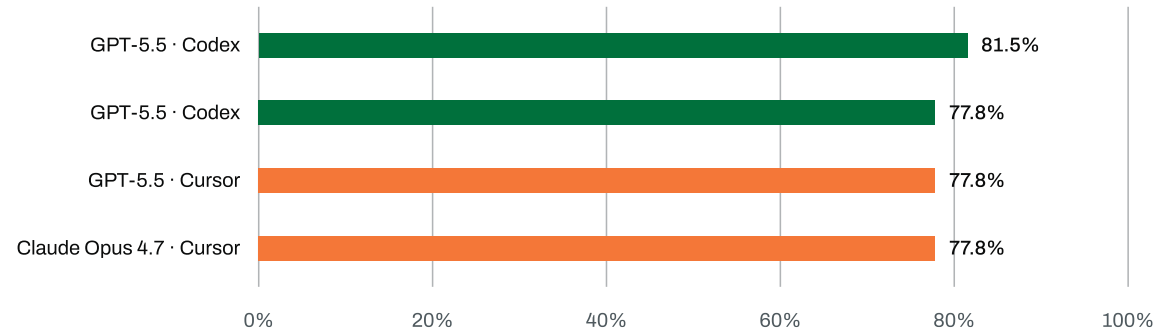
#### Limit

The author states that the 27-task pass-rate differences are not statistically significant.

Nyosegawa results

Dataset or repository

### Published results



# A-CODE-CLI repeated 10 web scenarios: OpenCode had a 2.4-point backend lead and cost 44% less than Claude Code.

## What they did

AIMultiple ran 10 full-stack web development scenarios 3 times with Claude Sonnet 4.6. Backend checks and Playwright tests measured correctness and user-interface behaviour.

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| Tasks           | 10                                                  |
| Repeated trials | yes                                                 |
| Grader          | Backend checks and Playwright user-interface checks |

## What we observe

OpenCode had a small backend lead over Claude Code and cost less in the matched Sonnet 4.6 comparison.

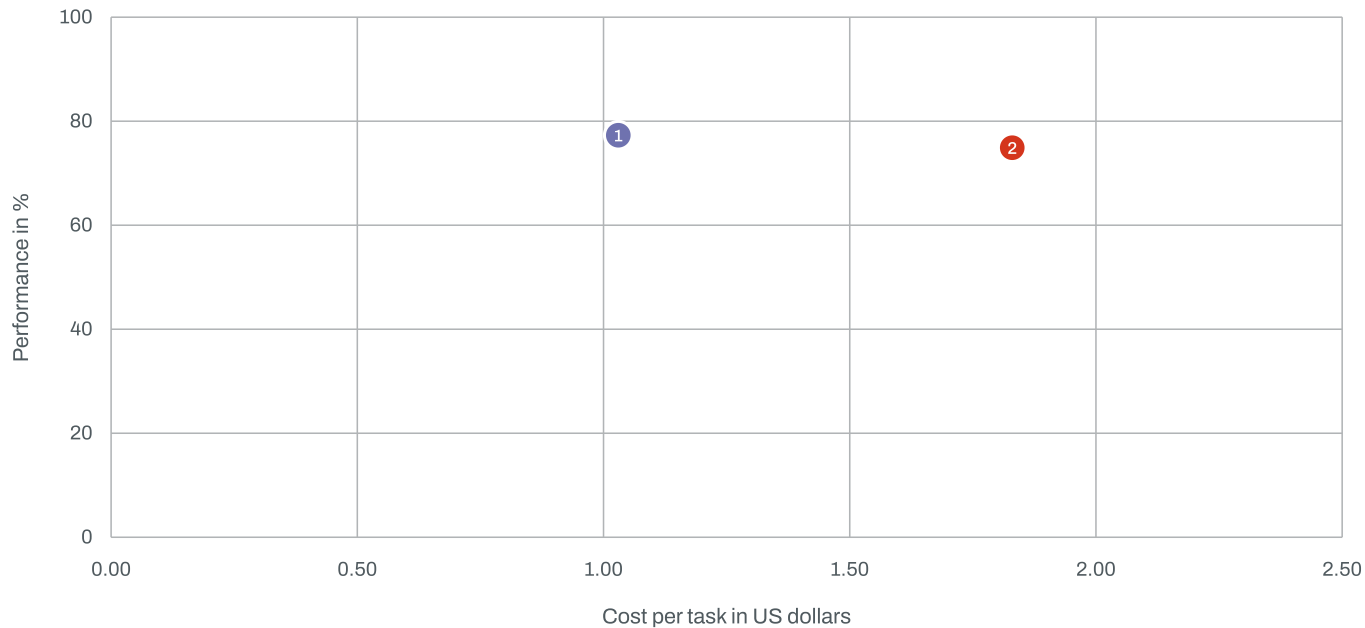
### Limit

The benchmark has only 10 scenarios. Only one example task is public.

### AIMultiple results

Dataset or repository

## Published results



| Point | Model             | Harness     | Result · cost  |
|-------|-------------------|-------------|----------------|
| 1     | Claude Sonnet 4.6 | OpenCode    | 77.3% · \$1.03 |
| 2     | Claude Sonnet 4.6 | Claude Code | 74.9% · \$1.83 |

## OpenBench repeated 3 coding tasks: every harness passed, so the useful difference was efficiency rather than correctness.

### What they did

OpenBench ran 3 repository coding tasks 3 times with GPT-5.5 at medium effort. Executable checkers awarded partial credit, while the study also recorded wall time and token use.

|                 |                                                 |
|-----------------|-------------------------------------------------|
| Tasks           | 3                                               |
| Repeated trials | yes                                             |
| Grader          | Executable checkers with partial-credit support |

### What we observe

All clean harnesses passed every task. Pi and Cursor used less time and fewer tokens.

#### Limit

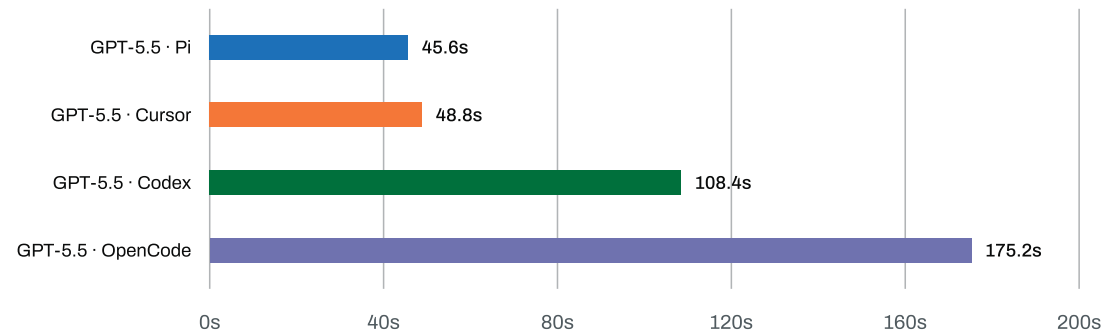
The tasks were too easy to separate frontier harnesses on correctness.

#### OpenBench results

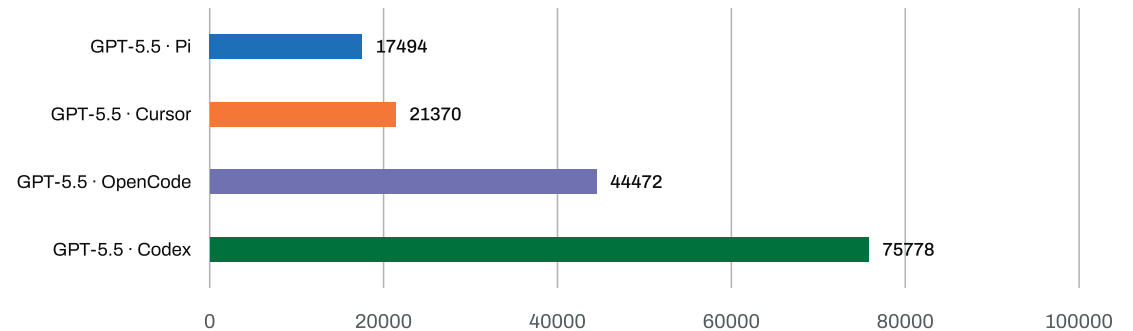
Dataset or repository

### Published results

#### Mean wall time



#### Tokens per solve



## Portkey measured 2 trivial requests: fixed input overhead ranged from about 2,600 tokens for Pi to 27,000 for Claude Code.

### What they did

Portkey routed 2 trivial prompts through its gateway and recorded the input payload sent by each harness. The exercise isolates fixed prompt overhead rather than coding performance.

|                 |                     |
|-----------------|---------------------|
| Tasks           | 2                   |
| Repeated trials | no                  |
| Grader          | Input-token capture |

### What we observe

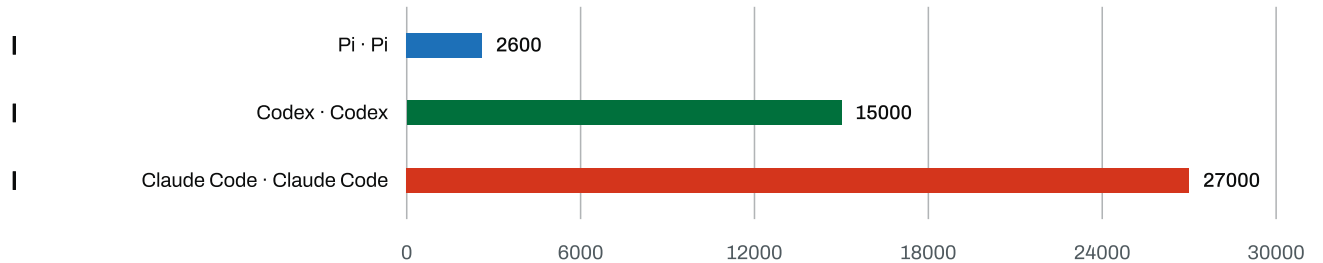
The fixed input payload ranged from about 2,600 tokens for Pi to 27,000 for Claude Code.

#### Limit

This is an overhead diagnostic. It does not measure coding quality or full-session economics.

Portkey results

### Published results



This diagnostic measures fixed input overhead. It does not measure coding quality.

# Evidence matrix

Twelve studies contain at least one matched-model harness comparison. Portkey is an efficiency diagnostic, not a quality benchmark.

| No. | Study                             | Task surface                                                                       | Model fixed | Includes Pi | Repeats   | Cost |
|-----|-----------------------------------|------------------------------------------------------------------------------------|-------------|-------------|-----------|------|
| 01  | Databricks coding-agent benchmark | Recent human-written pull requests from a multi-million-line internal codebase     | yes         | yes         | not clear | yes  |
| 02  | Warden benchmark                  | Known-vulnerability recognition in files tied to historical Sentry vulnerabilities | yes         | yes         | no        | yes  |
| 03  | Coding Agent Index                | 321 tasks across DeepSWE, Terminal-Bench v2 and SWE-Atlas-QnA                      | yes         | no          | yes       | yes  |
| 04  | DrugDiscoveryBench                | Expert-curated drug-discovery and life-sciences workflows                          | yes         | yes         | yes       | yes  |
| 05  | Agent Security League             | Real vulnerability-fixing tasks with hidden functional and security tests          | yes         | no          | no        | no   |
| 06  | scBench-Long                      | Long biological analysis workflows                                                 | yes         | yes         | yes       | no   |
| 07  | Vulnerability-Agent-Bench         | Real vulnerability repairs in isolated containers                                  | yes         | no          | no        | yes  |
| 08  | WildClawBench                     | Bilingual, multimodal and long-horizon agent tasks                                 | yes         | no          | not clear | yes  |
| 09  | CORE-Bench Hard                   | Scientific reproducibility tasks                                                   | yes         | no          | no        | yes  |
| 10  | HarnessBench                      | Recent bug-fix issues from 9 open-source repositories                              | yes         | no          | no        | yes  |
| 11  | A-CODE-CLI                        | Ten full-stack web development scenarios                                           | yes         | no          | yes       | yes  |
| 12  | OpenBench M4.5                    | Three repository coding tasks with executable checkers                             | yes         | yes         | yes       | no   |
| 13  | Harness Tax                       | Two trivial prompts routed through a gateway                                       | no          | yes         | no        | no   |

# Data and source access

External datasets remain under their publishers' terms. This repository stores derived observations and links to the original data.

| Source                                                                | Access                                                                             | Licence note                                                                               | Links          |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------|
| Benchmarking coding agents on Databricks' multi-million-line codebase | No public task-level dataset found                                                 | Not stated for benchmark data                                                              | Results        |
| Warden benchmark                                                      | Public aggregate table                                                             | Not stated for benchmark results                                                           | Results · Data |
| scBench-Long                                                          | Public JSON summary and public trajectories                                        | Check the linked benchmark repository and dataset terms                                    | Results · Data |
| OpenBench                                                             | Public repository with committed data for several milestones                       | MIT                                                                                        | Results · Data |
| DrugDiscoveryBench                                                    | Tasks and code are public; rubrics and reference answers are gated on Hugging Face | MIT for repository code and tasks; separate terms apply to bundled data and gated material | Results · Data |
| Artificial Analysis Coding Agent Index                                | Live public result page and methodology                                            | Not stated for result data                                                                 | Results        |
| Endor Labs Agent Security League                                      | Public leaderboard and articles                                                    | Not stated for benchmark data                                                              | Results        |
| Vulnerability-Agent-Bench                                             | Public explorer and linked Weights and Biases project                              | Not stated for benchmark data                                                              | Results · Data |
| WildClawBench                                                         | Public code, tasks, runtime images and dataset                                     | Check the repository and dataset cards                                                     | Results · Data |
| HarnessBench                                                          | Public code and results                                                            | Check the repository licence                                                               | Results · Data |
| CORE-Bench Hard                                                       | Public harness, leaderboard and encrypted traces                                   | Check the repository licence                                                               | Results · Data |
| A-CODE-CLI                                                            | Public report and one example task                                                 | Check the example repository licence                                                       | Results · Data |
| The harness tax                                                       | Public article with aggregate values                                               | Not stated for request logs                                                                | Results        |